

SONAM KARNA

A competent R & D and QC professional, targeting assignments as **QA/ QC Microbiologist in FMCG, Food, Beverage, Distillery, Water Treatment Plants, preferably in Odisha or West Bengal**

Profile Summary

- Highly skilled **R & D** and **QC** professional in the field of **Microbiological Research** with strong exposure of working for over 3 years on various Research Projects dealing in all phases such as **Sampling, Data Collection & Analysis, Result Computation** and so on
- Proven success in various **Microbiology Techniques & Processes** along with comprehensive exposure to **Micro Analysis Procedures, Microbiological Analysis, Microbiological Testing Laboratory** and all **QC/QA activities**
- **Strategic Researcher** with excellence in Quality Assurance, Quality Control, Project Management and implementing quality mechanisms to deliver desired output with focus on profitability by ensuring optimal utilization of resources
- Proficient in **WTP, STP/ ETP, Microbiological Tests, Biochemical Tests, Dairy Microbiology, etc.**
- **Successfully reviewed & updated Laboratory Standard Operating Procedures** in line with departmental objectives
- Ensured adherence to good documentation practices for **data integrity and traceability**

Areas of Expertise

Research & Development
Microbiology Operations
Quality Control/Assurance
Lab Operations
Sampling & Testing
Data Analysis
Safety/Risk Management
Regulatory Compliances/
SOPs
Project Management

Education

- M.Sc. in Applied Microbiology from KIIT University, Odisha, India in 2017
- B.Sc. in Microbiology from College of Basic Science and Humanities, Odisha, India in 2015

Certifications

- PGDCA in Computer Applications
- NGS Training by BDG Group

Technical Skills

- **Wet Lab Skills:** Molecular biology- Isolation of DNA, RNA, Plasmid, PCR Techniques, Gene Cloning Immunology- ELISA, Western Blotting, SDS PAGE, Immunoglobulin Purification by Affinity Column Chromatography
- **Microbiology:** Bacterial and Fungal Culture Techniques, Culture Preservation, Staining Techniques, Microscopy, Antibiotics Sensitivity, Biochemical Tests, Tissue Culture, Media Preparation
- **Cell Biology:** Microscopy (bright field, phase contrast, fluorescent, inverted), Live-Dead Cell Visualization using DAPI-PI, Giemsa Staining, Cell counting by Hemocytometer
- **Dry Lab Skills:** Bioinformatics Programming, Microsoft Excel, MYSQL, HTML, Linux, Pearl, MSA, Phylogenetic Analysis, NGS
- **Public Health:** MAXQDA Qualitative Data Analysis Tool, Community and MCH Research

Employment Details

Since Jan'21 | Research Assistant |
Regional Medical Research Center- ICMR, Bhubaneswar, India

Key Result Areas:

- Collecting samples and conducting analysis & test, issuing test reports as per documented procedure
- Ensuring compliance to safety procedures and regulations in the laboratory and highlighting any deviations
- Calibrating, maintaining & validating all laboratory equipment and instruments belonging to quality control laboratory in coordination with maintenance team
- Drafting & adhering to specifications, preparing and reviewing analytical method validation
- Maintaining laboratory records in accordance with Good Laboratory Practice standards
- Adhering to processes & SOPs, leveraging best practices to ensure process improvements, eliminate bottlenecks, realize operational efficiencies, control cost and reduce cycle time
- Performing inspections at various stages meeting safety & quality specifications without any deviation
- Fostering healthy working environment, mentoring & monitoring the performance of team members to ensure efficiency in process operations

Sep'20-Oct'20 | Medical Social Worker |
Regional Medical Research Center- ICMR, Bhubaneswar, India

Key Result Areas:

- Providing necessary support and assistance in completing research projects through necessary data collection, analysis, computing results and obtaining insights to be provided on the study
- Ensuring compliance to all regulations and SOPs while conducting research studies
- Maintaining accuracy, quality and confidentiality of data used in the research studies
- Prepared documentations and maintained all necessary records related to the research

Jan'19-Jun'19 | Project Lab Assistant |
National Rice Research Institute - ICAR, Cuttack, India

Projects & Publications

- Perception and determinants leading to antimicrobial (mis)use: A knowledge, attitude and practices study in the rural communities of Odisha, India with **Regional Medical Research Center- ICMR, Bhubaneswar, India**
- Study on challenges to access and utilization of maternal and child health services including immunization services during COVID 19 pandemic in Odisha and developing a context specific mitigation strategy applicable to similar health emergencies with **Regional Medical Research Center- ICMR, Bhubaneswar, India**
- Mental health and social stigma among healthcare personnel involved in the management of COVID 19 patients in India with **Regional Medical Research Center- ICMR, Bhubaneswar, India**
- Setting-up of a model biofertilizer production unit for supply of quality bio-inoculants for rice and rice based cropping system in Odisha with **National Rice Research Institute - ICAR, Cuttack, India**
- Bacteria causes community-acquired and nosocomial urinary tract infection in type-2 diabetes: A comparative approach with **School of Biotechnology- KIIT University, Bhubaneswar, India**
- Resolution of Resilience: Empirical findings on the challenges faced and the mitigation strategies adopted by community health workers (CHWs) to provide maternal and child health (MCH) services during the COVID 19 pandemic in the context of Odisha, India

Internship & Apprenticeship

Jun'17-Aug'18 | Apprentice Microbiologist | Denzong Brewery, Khordha, India
Nov'16-Jun'17 | Dissertation Trainee | School of Biotechnology- KIIT University, Bhubaneswar, India

Personal Details

Date of Birth: 24th March, 1995
Address: Bhubaneswar, India
Languages: English, Hindi and Odia